

Church numerals: $\underline{n} \stackrel{\text{def}}{=} \lambda f. \lambda x. \underbrace{f(\dots(f\ x)\dots)}_n$

Informally, a Church
numeral n means:
“to do something n times”

$$\underline{0} \stackrel{\text{def}}{=} \lambda f. \lambda x. x$$

$$\underline{1} \stackrel{\text{def}}{=} \lambda f. \lambda x. f\ x$$

$$\underline{2} \stackrel{\text{def}}{=} \lambda f. \lambda x. f\ (f\ x)$$

$$\underline{3} \stackrel{\text{def}}{=} \lambda f. \lambda x. f\ (f\ (f\ x))$$

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